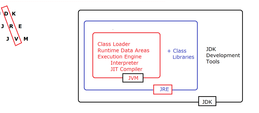
What is the difference between JDK, JRE,JVM and JIT compiler ?



JDK = JDK Development tools + JRE

JRE = Class Libraries + JVM

JVM = Class Loader + Runtime Data Areas + Execution Engine [Interpreter + JIT Compiler]

What is JDK :

* JDK stands for "Java Development Kit".

* It is a **developer version** that means by using JDK we can develop as well as execute our java programs.

* By using JDK we are able to execute our java code because internally, it contains JRE.

* In order to develop and execute java programs, JDK has provided the following development tools :

a) javac : Used to verify the syntax and convert the source code to byte code.

b) java : Java Launcher, used to load and execute .class file.

c) javap : Java Profiler, Used to get the complete information of a particular class OR interface.

d) jdb : Java debugger, Used to debug the code.

e) jconsole : Used to display the output on the console window.

f) javadoc : Java Documentation, Used to generate the java documentation

Note : All these tools are available as a part of JDK with following Path :

C:\> Programs file -> Java -> JDK 25 -> bin -> All tools

What is JRE :

* It stands for "Java Runtime Environment".

* It is a Client version so by using JRE, We can only execute (not develop) our java program.

* JRE is removed from JDK 11V because we introduced a new concept called JPMS (Java Platform Module System) available from JDK 9.

* So, from JDK 11V, We can write a java program and execute it directly, without compilation

```
Hello.java
**********
void main()
{
    IO.println("Hello Batch 60");
    IO.println("Welcome to Java 25");
}
```

Note : We can directly execute this program :

Command is : java FileName.java
Java Hello.java

What is JVM ?

* It stands for "Java Virtual Machine".

* It is used to load and execute the .class file.

* Along with this, It provides following features :

a) Loading the .class file.

b) Allocating the memory for the member of the class.

c) Memory management.

d) Deleting the un-used objects using Garbage Collector

e) Providing the security by using ByteCodeVerifier

f) Execute .class file line by line.

g) To improvise the performance of Java, Introduced JIT Compiler.

What is JIT Compiler ?

JIT (Just-In-Time) Compiler is a part of the JVM execution engine.

It is used to **enhance the performance** of Java programs.

Internally JVM uses an interpreter, which executes bytecode line by line hence execution is slow.

To overcome this limitation, JVM uses JIT Compiler.

How JIT compiler Works:

When the JVM finds some methods or instructions that are **repeated frequently** (called hotspots), Instead of interpreting them line by line again and again, JIT compiles those bytecode instructions into native machine code.

At the time of execution, JIT compiler directly provides these native machine code to JVM Interpreter for fast execution.

What is data type in java?

It defines the **type of data** that a variable can store.

It also describes:
a) Size -> How much memory it occupies in RAM.

b) Range -> What values it can hold (e.g., byte -> 128 to 127).

c) Operations -> What kind of operations can be performed (e.g., addition for numbers, concatenation for String).

How many data types in java?

Java data types are mainly divided into two categories:

1) Primitive Data type
2) Reference Data type [Never say non primitive data type]

Note : Out of 8 primary data types, 6 data types represents numbers (byte, short, int, long, float and double)

What is the difference between statically typed and dynamically typed language?

```
int x;  //Variable Declaration
x = 90; //Variable Initialization
```

Statically typed language:

The languages where data type is **compulsory** before initialization of a variable are called Statically typed language.
In these languages, once we define the **type of the variable** then it will hold same kind of value till the end of the program.

Example of statically typed languages:
C, C++, JSE, C# and so on

Dynamically typed language:

The languages where Data type **is not** compulsory, it is **optional** to initialize the variable are called Dynamically Typed Language.

In these languages we can provide all different kinds of values to the variable during the execution of the program.

Example of dynamically typed languages:
Visual Basic, JavaScript, Python and so on